Radian Measure and Arc Length

ANSWER KEY

Converting between degrees and radians

1 Convert each angle to radians, in exact form:

a $45° = \frac{\pi}{4}$ **b** $120° = \frac{2\pi}{3}$ **c** $270° = \frac{3\pi}{2}$ **d** $330° = \frac{11\pi}{6}$

2 Convert each angle to degrees. Round to one decimal place where needed:

a $\frac{\pi}{6} = 30°$ **b** $\frac{5\pi}{4} = 225°$ **c** $\frac{7\pi}{3} = 420°$ **d** $3.5\text{ rad} \approx 200.5°$

Arc length and sector area

3 A circle has radius 6 cm. Find the exact length of the arc subtended by a central angle of $\frac{2\pi}{3}$.
$s = r\theta = 6 \cdot \frac{2\pi}{3} = 4\pi \approx 12.57$ cm.

4 A sector of a circle of radius 10 cm has central angle $\frac{\pi}{5}$. Find its exact area.
$A = \frac{1}{2}r^2\theta = \frac{1}{2}(100)\frac{\pi}{5} = 10\pi \approx 31.42\text{ cm}^2$.

5 A wheel of radius 30 cm rolls along the ground without slipping.

a Through what angle, in radians, does the wheel turn when it travels 1.5 m? $\theta = \frac{s}{r} = \frac{150}{30} = 5$ radians.

b Express this angle to the nearest tenth of a degree. $5 \times \frac{180°}{\pi} \approx 286.5°$.

6 A pendulum of length 80 cm swings through an angle of $25°$. Find the length of the arc traced by its tip, to the nearest tenth of a centimetre.
$\theta = 25° = \frac{25\pi}{180}$ rad, so $s = 80 \cdot \frac{25\pi}{180} \approx 34.9$ cm.

Angular and linear speed

7 A carousel horse is 4 m from the center and completes one full rotation every 20 seconds.

a Find the angular speed of the horse, in radians per second. $\omega = \frac{2\pi}{20} = \frac{\pi}{10}$ rad/s.

b Find the linear speed of the horse, in m/s. $v = r\omega = 4 \cdot \frac{\pi}{10} = \frac{2\pi}{5} \approx 1.26$ m/s.

8 A car's wheel has radius 35 cm and spins at 400 revolutions per minute. Find the car's speed in km/h, to one decimal place.
$v = r\omega$, with $\omega = 400 \times 2\pi$ rad/min $= 800\pi$ rad/min. $v = 0.35 \times 800\pi \approx 879.6$ m/min $= 52{,}776$ m/h ≈ 52.8 km/h.

9 Two gears mesh together: the smaller gear has radius 5 cm and spins at 300 rpm. If the larger gear has radius 15 cm, find its rotational speed in rpm (the linear speed at the point of contact is the same for both gears).

$v = r_1\omega_1 = r_2\omega_2$: $5(300) = 15\,\omega_2$, so $\omega_2 = 100$ rpm.

Coterminal angles and reference angles

10 Find a positive coterminal angle (between 0 and 2π) for each:

a $\frac{9\pi}{4} \quad = \frac{\pi}{4}$ **b** $-\frac{\pi}{3} \quad = \frac{5\pi}{3}$ **c** $\frac{13\pi}{6} \quad = \frac{\pi}{6}$

11 Find the reference angle for each angle:

a $\frac{5\pi}{6} \quad = \frac{\pi}{6}$ **b** $\frac{7\pi}{4} \quad = \frac{\pi}{4}$ **c** $\frac{4\pi}{3} \quad = \frac{\pi}{3}$

The radian definition itself

12 State the definition of one radian in terms of arc length and radius, and explain why radians are considered a 'natural' unit for angles.

One radian is the angle subtended at the center of a circle by an arc whose length equals the radius. Radians are natural because the arc length formula $s = r\theta$ has no extra conversion constant when θ is in radians, unlike degrees.

13 Explain why the full circle corresponds to 2π radians, using the circumference formula $C = 2\pi r$.

Since $s = r\theta$ and the full circumference is $s = 2\pi r$, setting $r\theta = 2\pi r$ gives $\theta = 2\pi$ — the full circle sweeps exactly 2π radians.

Combined arc length and sector area problems

14 A sprinkler sprays water over a sector of radius 8 m and central angle $\frac{\pi}{3}$. Find both the area watered and the length of the outer arc of the watered region.

Area: $A = \frac{1}{2}(8)^2\left(\frac{\pi}{3}\right) = \frac{32\pi}{3} \approx 33.5$ m^2. Arc length: $s = 8\left(\frac{\pi}{3}\right) = \frac{8\pi}{3} \approx 8.4$ m.

15 A slice of pizza (a sector) has radius 12 cm and arc length 10 cm along its crust. Find the central angle in radians and the area of the slice.

$\theta = \frac{s}{r} = \frac{10}{12} = \frac{5}{6}$ rad. Area $= \frac{1}{2}r^2\theta = \frac{1}{2}(144)\left(\frac{5}{6}\right) = 60$ cm^2.

Common angles in both units (reference table)

16 Complete a conversion table for the common angles $0°, 30°, 45°, 60°, 90°, 180°, 360°$ in radians.

$0° = 0$; $30° = \frac{\pi}{6}$; $45° = \frac{\pi}{4}$; $60° = \frac{\pi}{3}$; $90° = \frac{\pi}{2}$; $180° = \pi$; $360° = 2\pi$.

Satellite and orbital speed applications

17 A satellite orbits Earth at a radius of 7000 km, completing one orbit every 2 hours. Find its angular speed in rad/h and its linear speed in km/h, to the nearest whole number.

$\omega = \frac{2\pi}{2} = \pi$ rad/h. $v = r\omega = 7000\pi \approx 21{,}991$ km/h.

18 Earth rotates once every 24 hours. Find the linear speed (due to rotation alone) of a point on the equator, using Earth's radius ≈ 6371 km, to the nearest km/h.

$\omega = \frac{2\pi}{24} = \frac{\pi}{12}$ rad/h. $v = 6371 \cdot \frac{\pi}{12} \approx 1668$ km/h.

Negative angles and full-turn arcs

19 Find the exact arc length swept by a radius-4 circle for a rotation of $-\frac{3\pi}{2}$ radians (clockwise), and interpret the sign.

Arc length uses the magnitude of θ: $s = 4\left(\frac{3\pi}{2}\right) = 6\pi$. The negative sign in the angle indicates clockwise rotation, but arc length itself is always a positive (unsigned) distance.